



- 8 The following table shows part of the instruction set for a processor. The processor has two registers, the Accumulator (ACC) and the Index Register (IX).

| Instruction                                                                                                                                                                         |            | Explanation                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Opcode                                                                                                                                                                              | Operand    |                                                                                                                                               |
| LDM                                                                                                                                                                                 | #n         | Immediate addressing. Load the number n to ACC                                                                                                |
| LDD                                                                                                                                                                                 | <address>  | Direct addressing. Load the contents of the location at the given address to ACC                                                              |
| LDI                                                                                                                                                                                 | <address>  | Indirect addressing. The address to be used is at the given address.<br>Load the contents of this second address to ACC                       |
| LDX                                                                                                                                                                                 | <address>  | Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC |
| LDR                                                                                                                                                                                 | #n         | Immediate addressing. Load the number n to IX                                                                                                 |
| ADD                                                                                                                                                                                 | #n/Bn/&n   | Add the number n to the ACC                                                                                                                   |
| ADD                                                                                                                                                                                 | <address>  | Add the contents of the given address to the ACC                                                                                              |
| SUB                                                                                                                                                                                 | #n/Bn/&n   | Subtract the number n from the ACC                                                                                                            |
| SUB                                                                                                                                                                                 | <address>  | Subtract the contents of the given address from the ACC                                                                                       |
| INC                                                                                                                                                                                 | <register> | Add 1 to the contents of the register (ACC or IX)                                                                                             |
| DEC                                                                                                                                                                                 | <register> | Subtract 1 from the contents of the register (ACC or IX)                                                                                      |
| <address> can be an absolute or a symbolic address<br># denotes a denary number, e.g. #123<br>B denotes a binary number, e.g. B01001010<br>& denotes a hexadecimal number, e.g. &4A |            |                                                                                                                                               |

